Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

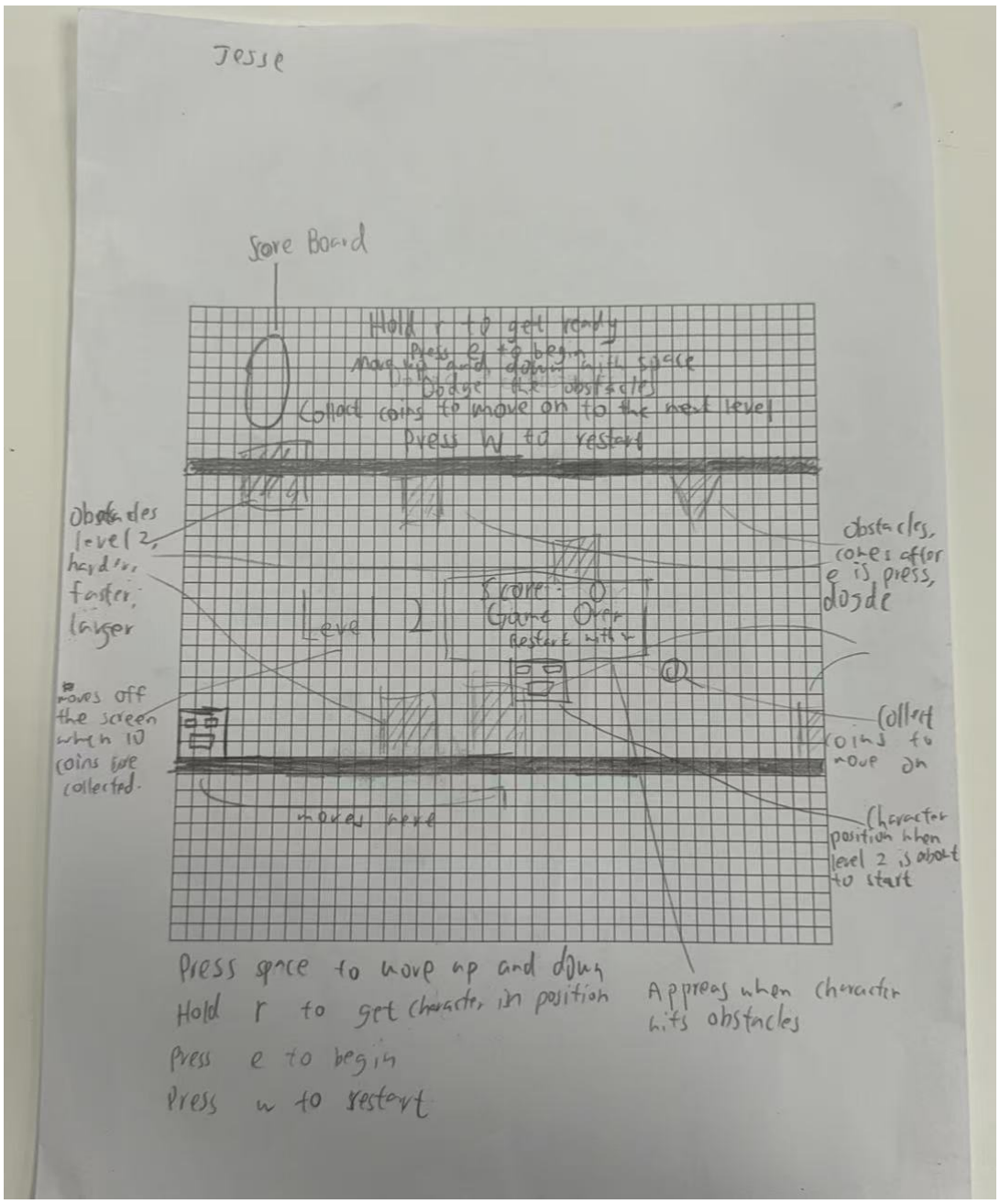
My program is a two-level obstacle-dodging game called "Sky Jumper." The player controls a character that can move vertically by pressing the spacebar to toggle direction. The goal is to avoid obstacles while collecting yellow coins. After collecting 10 coins in Level 1, the game transitions to Level 2 with faster, red-colored obstacles. The game includes a "get ready" thing where players must hold 'r' to move their character to the starting position, then press 'e' to begin. If the player hits an obstacle, the game ends and appears a Game Over screen with their score.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)



(400,400)

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
Groups	To organize multiple shapes into one object. This allows me to move, hide, and check if hits for complex objects as one unit.

onStep	To animate a shape, create movements for shapes, and make the game for realistic and fun.
Custom Properties	To track game state, these properties control movement speed.
onKeyHold/onKeyPress	To do something when the key is pressed or hold, hold “r” to get ready, “e” to begin, “w” to restart, “space” to move.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

- Describe the Problem: What specific issue did you encounter?
  The issue that I encountered was when I wanted to transition from level 1 to level 2, and I didn’t know how to do that, I needed to do a lot of things for it to be what I imagined. Like stop all level 1 obstacles from moving, hide level 1 obstacles, show level 2 transitioning sign, then show level 2 obstacles...
- Attempted Solutions: What did you try in order to solve the problem?
  I first tried just changing the level and switching obstacles immediately, but the sudden change was confusing for players, and level 2 is much faster. It will be very difficult for the players to transition.
- Final Solution: What was the final solution that fixed the issue?
  The final solution was, after 10 coins is collected, the player stays slightly down, and a level 2 sign appear and moves off the screen. After all this is done, the level 2, harder obstacles come, and when you press the spacebar, the game goes back to how it goes.

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Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
Groups	45 - 70	A group combines multiple shapes into one object, which can be moved or changed as a whole	I made obstacles 1,2,3,4 having multiple shapes in them, and I put them in onKeyPress, functions, onStep, to make them move as a whole in the way I wanted.
onKeyPress	139 - 157	To run a specific action when the key is pressed.	For me, pressing "e" means to start the game, to have the obstacles start moving. Pressing "space" means to control the character's movement vertically. Pressing "w" means to restart the game when it's over.
onKeyHold	160 - 163	To run a specific action when the key is hold.	Holding "r" moves the character to its ready location. Hold it completely until it can't move.
onStep	165 - 279	To animate a shape, so it appears to be moving.	This is a big part of my code, it is how everything moves. The obstacles starts wrapping around, the character moves, the level 2 sign moves, the level 2 obstacles come and starts wrapping around, coins appear in the right times.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?

- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

Code Changes: I think I'll change the code so that coins won't appear inside the obstacles and won't be impossible to claim. I would also like to add a level 3, even a level 4, and a winning animation.
Process Adjustments: I didn't really stick to my original plan because at first, I wasn't exactly sure what I was going to do. I think these changes were beneficial because they really led me to a great program, where I could even add more into the game. To improve my planning, I should still have the planning 100% done before moving on to coding.
Resource Utilization: I think I did a good job on using available resources, I used class notes and cmu academy's concepts we learned. I asked peers for some possible ideas or improvements. Some additional resources that could have been beneficial is probably asking the teacher more.
Time Management: After the planning, I did a good job on time management, I did the work that I'm supposed to do in class and at home. I still would spend a bit more time on planning before I start coding.